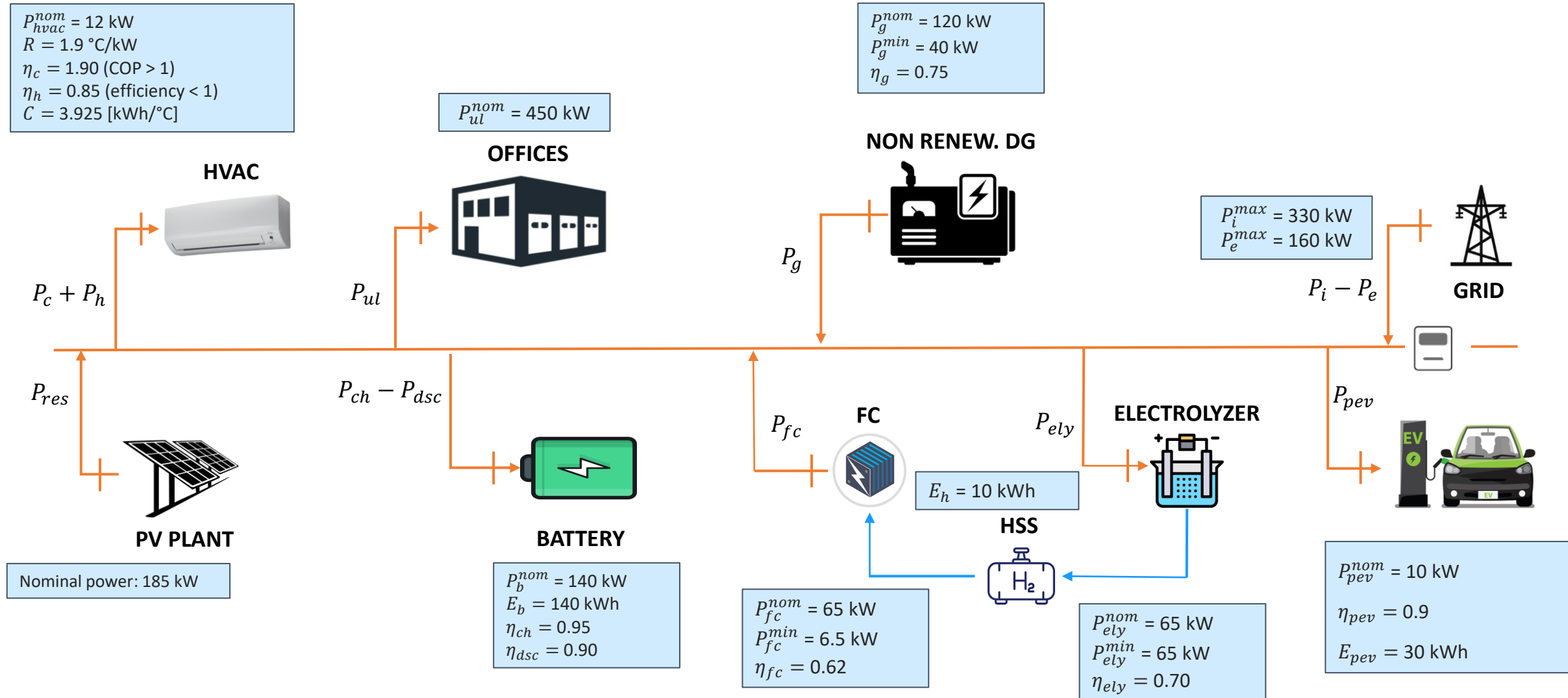


Project 13



Project 9



Control specifications

Objectives:

- minimize the cost / maximize the income of the industrial district;
- keep the temperature around the set-point $T^{sp} = 22^{\circ}\text{C}$ (from April to Sept.) $T^{sp} = 20^{\circ}\text{C}$ (from Oct. to March) and with a tolerance of $\Delta = 2^{\circ}\text{C}$;
- keep the battery State-of-Charge (SoC) within the interval $[0.1 \ 0.9]$;
- recharge the PEV when required

Assumptions:

- P_c e P_h can be set in the interval $[0 \ P_{hvac}^{nom}]$;
- $\Delta t_s = 1 \text{ h}$;
- $T(k)$, $T_{ex}(k)$, $P_{pv}(k)$, $SoC(k)$, $P_{ul}(k)$, are measured at time k ;
- at time k we are provided with the forecasts T_{ex} , P_{pv} , and P_{ul} for the next 24h;
- cost of imported energy $c_l(k)$ is known for the next 24h (see next slide);
- price of exported energy $p_e(k)$ is known for the next 24h (we use PUN);
- price of fuel for the non renewable DG c_f is constant and known.

Project 9



Dataset:

- T_{ex} : T_ex_rome_campus_bio_medico_2022.mat
- P_{pv} : lr_rome_campus_bio_medico_2022.mat
- P_{ul} : office_load.mat (scale the value to obtain the right nominal value)
- c_l : F1 = 0,53276 F2=0,54858 F3=0,46868 [€/kWh]
- p_e : PUN_2022.mat
- c_f : test with 0.15 and with 0.45 €/kWh